

Planar Turán Numbers of Triangular Blocks

January 6, 2026

Abstract

The planar Turán number $\text{ex}_{\mathcal{P}}(n, H)$ is the maximum number of edges in an n -vertex planar graph not containing H as a subgraph. In this paper, we consider the planar Turán numbers of graphs which we call triangular blocks. This family was already implicit in the literature on planar Turán numbers via the “contribution method” introduced by Ghosh, Györi, Martin, Paulos and Xiao in [3]. Using the contribution method, we estimate the planar Turán numbers of a collection of triangular blocks which we call isopods and write $I_4, I_5, I_6, I_7, I_5^1, I_5^2$. We also provide extremal constructions which meet these estimates for infinitely many n . Lastly, by identifying plane triangulations with particular properties, we show that all but finitely many triangular blocks have planar Turán number $3n - 6$ for all n .

Contents

1	Introduction and main results	1
2	The contribution method	5
3	I_4, I_5, I_6, I_7	7
4	$I_5^1, I_5^2, I_6^1, I_6^2$ [In Progress]	8
5	Extremal constructions	10
6	$\text{ex}_{\mathcal{P}}(n, B) = 3n - 6$	12

1 Introduction and main results

In this paper, all graphs considered are planar, undirected, and finite and contain neither loops nor multiple edges.

A beautiful theorem in extremal graph theory is Turán’s Theorem which gives the maximum number of edges that an n vertex graph can have without containing a K_r subgraph. The general Turán type extremal graph theory problem asks how many edges an n vertex graph can have while

avoiding a particular subgraph H . This number is called the Turán number or extremal number of H and is denoted $\text{ex}(n, H)$.

Dowden 2016 [1] initiated the study of Turán type problems where the host graph is planar. If H is a graph, the planar Turán number of H , denoted $\text{ex}_{\mathcal{P}}(H, n)$, is defined to be the maximum number of edges possible in a planar graph on n vertices which does not contain H as a subgraph. Previous work has found the Planar Turán numbers for various cycles ([1],[3],[6], [4]), cycles with chords [7], friendship graphs [2], wheels and stars [5], and other graphs. Ghosh-Győri-Xiao [7] introduced the notion of a triangular block. In that paper, triangular blocks are the classes for a particular partition of the edge set $E(G)$ of a graph G . For us, triangular blocks are a class of graphs induced by these partition classes along with a labeling ϕ of their inner and outer faces.

Definition 1.1 (Face connected). Let G be a planar graph and $\phi : F(G) \rightarrow \{0, 1\}$ be a function. Then, fix an edge $e \in G$ and let $\Omega_0 = \{e\}$. Define

$$\Omega_{i+1} = \Omega_i \cup \{\text{edges sharing a face } f \text{ with some edge in } \Omega_i \text{ such that } \phi(f) = 1\}$$

Say G is ϕ -face connected if, given any starting edge in G , we have $E(G) = \bigcup_{i=0}^{\infty} \Omega_i$.

If $\phi(f) = 1$, then we call f an inner face. Otherwise, $\phi(f) = 0$ and we call f an outer face. Edges which border at least one outer face are called outer edges and edges bordering no outer faces are called inner edges. By Definition 1.1, a graph is ϕ -face connected if the graph of adjacent inner faces is a connected graph.

Definition 1.2 (Triangular block). A triangular block is a planar graph B along with a function $\phi : F(B) \rightarrow \{0, 1\}$ satisfying the following properties

1. If $f \in F(B)$ is a k -face, $k > 3$, then $\phi(f) = 0$;
2. B is ϕ -face connected.

In a triangular block B , all k -faces with $k > 4$ are outer faces. However, it is possible that some 3-faces are outer faces, so long as there are enough 3-faces are inner for B to be ϕ -face connected. Any graph can be partitioned into ϕ -connected components which are triangular blocks, a procedure described in [7]. The following definition make this precise.

Definition 1.3. Fix a planar graph G . For $e \in G$, let $\Omega_0^e = \{e\}$ and define

$$\Omega_{i+1}^e = \Omega_i^e \cup \{\text{edges sharing a 3-face } f \text{ with some edge in } \Omega_i^e\}$$

Then, $[e] = \bigcup_{i=0}^{\infty} \Omega_i^e$ is the partition class of e . Each partition class $[e]$ gives rise to a graph $G_{[e]}$ by removing edges not in $[e]$. Any 3-face in G for which all the edges are in $[e]$ becomes a 3-face in $G_{[e]}$; we say that such a face is inherited from G . The planar graph $G_{[e]}$ is a triangular blocks with respect to the labeling $\phi_G : F(B) \rightarrow \{0, 1\}$ given by

$$\phi_G(f) = \begin{cases} 1 & f \text{ is inherited from a 3-face in } G \\ 0 & \text{otherwise} \end{cases}.$$

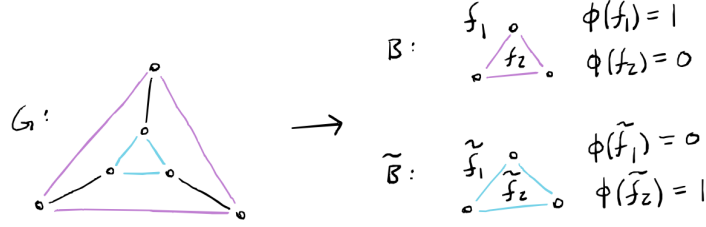


Figure 1: We partition G into two triangular blocks.

The inherited/interior faces are the only “real” faces in $G_{[e]}$ in the sense that any non-inherited/exterior face might not remain a single face when embedded in G .

Example 1.4. Let G be the graph on the left of Figure 1 and let

$$\phi_G(f) = \begin{cases} 1 & f \text{ is inherited from a 3-face in } G \\ 0 & \text{otherwise} \end{cases}.$$

Then, G has two ϕ -connected components and it splits into two triangular blocks B and $\tilde{B}$ as shown in the same figure. Notice that the 4-faces in G are no longer detectable in the representation as triangular blocks B and $\tilde{B}$.

The map ϕ_G allows us to “forget” the graph G a triangular block (B, ϕ_G) comes from and still retain important information about how it fits into G .

Definition 1.5. Given a planar graph G , let $\mathcal{T}(G) = \{G_{[e]} \mid e \in E(G)\}$ be the collection of all triangular blocks obtained by partitioning $E(G)$ in the manner of Definition 1.3.

We now describe families of finite graphs which can be obtained as triangular blocks. For these graphs, we understand the infinite face to be the unique outer face.

Definition 1.6. We define the following families.

1. The d -wheel W_d is the planar graph consisting of a $d - 1$ cycle with all its vertices connected to a single central vertex.
2. The d -isopod I_d is the planar graph obtained from W_d by deleting an outer edge.¹
3. Index the $k - 2$ faces of I_d in order from 1 to d from left to right. The isopod with a hat I_d^i is the planar graph obtained from I_d by gluing a triangle adjacent to the i th face.
4. Let $J_{i,j}$ denote the planar graph on vertices $\{1, 2, \dots, i + j - 2, x, y\}$ with edges $\{n, n + 1\}$ for $1 \leq n < i + j - 2$, edges $\{x, n\}$ for $n = 1, \dots, i$, and edges $\{y, n\}$ for $n = i - 1, \dots, i + j - 2$.

¹We call d -isopods after the crustaceans in the order *Isopoda* which the authors believe resemble the graphs in question.

Planar triangular numbers for wheels were found in [5]. d -isopods are the next family of planar triangular blocks which we study and we calculate their planar triangular numbers for all d .

Our main results are can be broken up as (1) obtaining planar Turán numbers less than $3n - 6$ for graphs $I_4, I_5, I_6, I_7, I_5^1, I_5^2$ and (2) showing that all but finitely many other triangular blocks B have planar Turán number $\text{exp}_{\mathcal{P}}(B, n) = 3n - 6$ for all n and explicitly enumerating those triangular blocks where the planar Turán number remains unknown.

Theorem 1.7. *For sufficiently large n , we have*

1. $\text{exp}_{\mathcal{P}}(n, I_4) \leq \frac{12}{5}(n - 2);$
2. $\text{exp}_{\mathcal{P}}(n, I_5) \leq \frac{8}{3}(n - 2);$
3. $\text{exp}_{\mathcal{P}}(n, I_6) \leq \frac{48}{17}(n - 2);$
4. $\text{exp}_{\mathcal{P}}(n, I_7) \leq \frac{120}{41}(n - 2);$
5. $\text{exp}_{\mathcal{P}}(n, I_5^1) \leq \frac{36}{13}(n - 2);$
6. $\text{exp}_{\mathcal{P}}(n, I_5^2) \leq \frac{36}{13}(n - 2);$

All of the above bounds have equality for infinitely many n .

The proofs for these bounds are given in section 3 and constructions meeting these bounds for infinitely many n are given in section 5.

We can now list the triangular blocks whose planar Turán numbers are less than $3n - 6$ or remain unknown to the authors.

Theorem 1.8. *Suppose that B is a triangular block with planar Turán number $\text{exp}_{\mathcal{P}}(B, n) < 3n - 6$ for some n . Then, B belongs to the following list.*

- Two vertices connected by a single edge;
- The complete graph K_3 ;
- $I_4, I_5 = I_4^1, I_6, I_7$;
- I_5^1, I_5^2 ;
- $I_6^1, I_6^2, I_7^1, I_7^2, I_7^3$;
- $J_{3,4}, J_{4,4}, J_{3,5}, J_{4,5}, J_{3,6}$.

The graphs for which we calculate planar Turán numbers less than $3n - 6$ are those given in the first three lines above and listed in Theorem 1.7.

Our paper consists of four main sections. Section 2 provides essential background on a technique for bounding Planar Turán numbers called the contribution method. Section 3 applies the contribution method to bound the planar Turán numbers of the graphs in Theorem 1.7. Section 5 gives extremal constructions which meet the bounds in the previous section. Section 6 proves 1.8 by showing that all but finitely many triangular blocks are avoided by one of two plane triangulations.

2 The contribution method

Face and edge contribution methods were introduced in [3]. The contribution method allows us to find upper bounds for the quantities $\text{ex}_{\mathcal{P}}(n, G)$ in terms of local information about triangular blocks B in G . In this section, we show how these bounds can be obtained by thinking about (B, ϕ_G) without knowing the embedding of B into G .

Let G be a planar graph and let $B \subseteq G$ with inner/outer faces assigned by $\phi_G : F(B) \rightarrow \{0, 1\}$. We introduce the following notation:

$e(G)$ the number of edges in G ;

$f(G)$ the number of faces in G ;

n_B the number of vertices in B ;

e_B the number of edges in B ;

f_B the number of interior faces of B ;

k_B the number of exterior faces of B ;

s_B the number of exterior edges of B .

Definition 2.1. An edge $e \in E(B)$ is contained in either one or two exterior faces. If e is contained in two exterior faces, then B must be a copy of K_2 . In this case, B touches two outer faces in G . Call these faces f_1 and f_2 . Then, the *face contribution* of B is defined to be $\text{fc}(B) = \frac{1}{|f_1|} + \frac{1}{|f_2|}$. If B is not K_2 , then for every outer edge e , let $f_B(e)$ equal the size of the outer face that contains e in G . Then, define the *face contribution* of B to be

$$\text{fc}_B = f_B + \sum_{\substack{e \text{ outer} \\ \text{edge of } B}} \text{fc}_B(e). \quad (1)$$

Consider the collection $\mathcal{T}(G)$ of all triangular blocks in G .

Proposition 2.2.

$$e(G) = \sum_{B \in \mathcal{T}(G)} e_B.$$

$$f(G) = \sum_{B \in \mathcal{T}(G)} \text{fc}_B.$$

Proof. The triangular blocks are edge-disjoint by definition and each edge in G is included in some triangular block. Therefore,

$$e(G) = \sum_{B \in \mathcal{T}(G)} e_B.$$

For the face contribution, note that the blocks also partition the boundary edges of each face $f \in F(G)$. Thus,

$$\sum_{B \in \mathcal{T}(G)} \text{fc}_B = \sum_{B \in \mathcal{T}(G)} \sum_{f \in F(G)} \frac{|E(f) \cap E(B)|}{|f|} = \sum_{f \in F(G)} \sum_{B \in \mathcal{T}(G)} \frac{|E(f) \cap E(B)|}{|f|}.$$

Since the blocks partition $E(f)$, we have $\sum_{B \in \mathcal{T}(G)} |E(f) \cap E(B)| = |E(f)|$. Therefore,

$$\sum_{B \in \mathcal{T}(G)} \text{fc}_B = \sum_{f \in F(G)} \frac{|E(f)|}{|f|} = \sum_{f \in F(G)} 1 = f(G).$$

□

Definition 2.3. Given a triangular block $B \subseteq G$ for some planar graph G , define the ratio

$$q_B = \frac{e_B}{e_B - \text{fc}_B}.$$

Lemma 2.4. *If G is a planar graph, then $\exp(n, G) \leq \sup_{G \not\supseteq B} q_B(n-2)$ where the supremum is taken over all triangular blocks B avoiding G as a subgraph.*

Proof. Fix a planar graph G and let A be a planar graph on n vertices avoiding G . Let $Q = \sup_{B \not\supseteq G} q_B$. Then, for any triangular block B a subgraph of A , we have $q_B = \frac{e_B}{e_B - \text{fc}_B} \leq Q$. So, $e_B - Q(e_B - \text{fc}_B) \leq 0$. Summing over triangular blocks B which are subgraphs of A , we have $0 \geq \sum_{B \in \mathcal{T}(A)} (e_B - Q(e_B - \text{fc}_B)) = e(A) - Q(e(A) - f(A))$. Rewriting this with Euler's formula shows that $e(A) \leq Q(n-2)$. Since A was chosen arbitrarily, so $\exp(n, G) \leq Q(n-2)$. □

Lemma 2.5. *Given a triangular block with B ,*

$$\begin{aligned} e_B &= 3n_B + 3k_B - s_B - 6; \\ f_B &= 2n_B + 2k_B - s_B - 4. \end{aligned}$$

Proof. By Euler's formula,

$$n_B - e_B + f_B + k_B = 2.$$

Moreover, we can count the number of incidences between interior faces and edges in two ways: every internal edge is incident to 2 internal faces, every external edge is incident to 1 internal face; whereas every internal face is incident to 3 edges.

$$2(e_B - s_B) + s_B = 3f_B.$$

Solving this system of equations yields the desired equality. □

Lemma 2.6. *Let B be a nontrivial triangular block of a planar graph G that is not G itself, then*

$$q_B \leq \frac{3(n_B + k_B - 2) - s_B}{n_B + k_B - 2 - \frac{s_B}{4}}.$$

Proof. Let B be such a triangular block. Every exterior face of B is incident to at least 4 edges, so $\text{fc}(B) \leq \frac{s_B}{4} + f = 2n_B + 2k_B - \frac{3s_B}{4} - 4$. Therefore, $q_B = \frac{e_B}{e_B - \text{fc}_B} \leq \frac{3n_B + 3k_B - s_B - 6}{n_B + k_B - 2 - \frac{s_B}{4}}$. □

For fixed n , this bound is achieved when all of the exterior faces of B are 4-faces.

3 I_4, I_5, I_6, I_7

In this section, we prove a bound for the planar Turán numbers of the graphs I_4, I_5, I_6, I_7 using the contribution method. Throughout the section, B will denote a triangular block with n_B vertices, e_B edges, f_B interior faces, k_B exterior faces, and s_B exterior edges. Our first lemma bounds the total degree of any triangular block avoiding I_d in terms of the number of exterior faces.

Lemma 3.1. *Given a triangular block B with k_B exterior faces, if it avoids the isopod I_d as a subgraph, then*

$$\sum_{x \in V(B)} \deg(x) \leq (d-2)(k_B + n_B - 1).$$

Proof. We induct on k_B , the number of exterior faces.

($k_B = 1$) If B has ≤ 1 exterior faces, then there cannot be a vertex of degree $\geq d-1$ or there would be an I_d induced by the high-degree vertex and its neighbors. So, $\sum_{x \in V(B)} \deg(x) \leq (d-2)n_B$.

($k_B > 1$) Assume the statement is true for a block with k exterior faces. Let B be a triangular block on n vertices with $k_B = k + 1$ outer faces and at least vertex x of degree $\geq d-1$. In order to avoid I_d , it must be the case that x is on at least two outer faces. We split these two outer faces by replacing x with x_1 connected like x to the first outer face and x_2 connected to the remaining neighbors of x . This gives us B' with $k_{B'} = k_B - 1 = k$ outer faces and $n_{B'} = n_B + 1$ vertices. By the induction hypothesis,

$$\sum_{x \in V(B')} \deg(x) \leq (d-2)(k_{B'} + n_{B'} - 1).$$

Since the sum of degree of B and B' are the same, we have

$$\sum_{x \in V(B)} \deg(x) \leq (d-2)(k_B + n_B - 1)$$

which completes the proof. $\square$

In particular, since the total degree of a graph is twice the number of edges, so this tells us $2e_B \leq (d-2)(k_B + n_B - 1)$. We can then bound q_B (Definition 2.3) using this bound and the identities in Lemma 2.5.

Lemma 3.2. *We have $q_B \leq \frac{4n_B(d-2)}{dn_B-2}$ for B a triangular block avoiding I_d .*

Proof. By Lemma 3.1 and Proposition 2.5, $2(3n_B + 3k_B - s_B - 6) = 2e_B = \sum_{x \in V} \deg(x) \leq (d-2)(k_B + n_B - 1)$. Solving for s_B , $2s_B \geq (8-d)n_B + (8-d)k_B + d - 14$. By Lemma 2.6,

$$\begin{aligned} q(B) &\leq \frac{3(n_B + k_B - 2) - s_B}{n_B + k_B - 2 - \frac{s_B}{4}} \\ &\leq \frac{24(n_B + k_B - 2) - 4((8-d)n_B + (8-d)k_B + d - 14)}{8(n_B + k_B - 2) - ((8-d)n_B + (8-d)k_B + d - 14)} \\ &= \frac{4(d-2)(n_B + k_B - 1)}{d(n_B + k_B - 1) - 2} \leq \frac{4n_B(d-2)}{dn_B - 2}. \end{aligned}$$

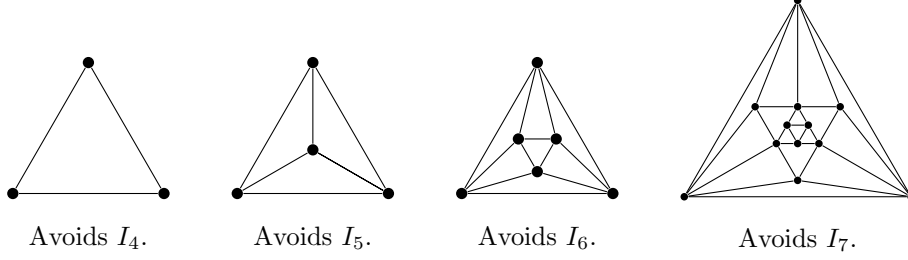


Figure 2: Triangulations avoiding I_4, I_5, I_6, I_7 .

□

Remark 3.3. Equality holds in Lemma 3.2 if $k_B = 1$ and $s_B = 4$, i.e. B has a single exterior face with 4 edges.

This immediately provides us the bounds for $\exp(n, I_d)$ given in Theorem 1.7.

Proof of I_d bounds in Theorem 1.7. We apply Lemma 3.2 to each I_d , $4 \leq d \leq 7$.

1. If B avoids I_4 , then $q_B \leq \frac{4n_B}{2n_B-1}$. If $n \geq 3$, then $\frac{4n_B}{2n_B-1} \leq \frac{12}{5}$.
2. If B avoids I_5 , then $q_B \leq \frac{12n_B}{5n_B-2}$. If $n \geq 4$, then $\frac{12n_B}{5n_B-2} \leq \frac{8}{3}$.
3. If B avoids I_6 , then $q_B \leq \frac{8n_B}{3n_B-1}$. If $n \geq 6$, then $\frac{8n_B}{3n_B-1} \leq \frac{48}{17}$.
4. If B avoids I_7 , then $q_B \leq \frac{20n_B}{7n_B-2}$. If $n \geq 12$, then $\frac{20n_B}{7n_B-2} \leq \frac{120}{41}$.

By Lemma 2.4, $\exp(n, I_d) \leq \sup_{I_d \not\subseteq B} q_B(n-2)$ and this completes the proof of the bounds. □

Remark 3.4. The question arises whether the bounds on $\sup_{I_d \not\subseteq B} q_B$ are tight. For I_d fixed, choose T to be the plane triangulation avoiding I_d in Figure 2. One can verify that q_T equals the upper bound given in the above proof. This means, for instance, that $\sup_{I_4 \not\subseteq B} q_B = \frac{12}{5}$ and likewise for the other suprema.

Remark 3.5. For $d = 8$, the bound in Lemma 3.2 looks like $q_B \leq \frac{24n}{8n-2}$. However, since $\frac{24}{8n-2} \geq 3$, so the bound this gives for $\exp(n, I_8)$ is no better than $\exp(n, I_8) \leq 3(n-2) = 3n-6$ which is the universal bound for the number of edges in a planar graph. This demonstrates how I_8 is the natural limit to this bound and, indeed, graphs I_d with $d \geq 8$ have planar Turán number $3n-6$ as we will demonstrate in Section 6.

4 $I_5^1, I_5^2, I_6^1, I_6^2$ [In Progress]

Let G be a planar graph. A fractional edge partition of G is a family of functions $\kappa_i : E(G) \rightarrow [0, 1]$ over some indexing set I such that for each $e \in E(G)$, we have $\sum_{i \in I} \kappa_i(e) = 1$. Observe that to each κ_i , we can associate the subgraph G_i with edge set $\text{supp } \kappa_i = \{e \in E(G) \mid \kappa_i(e) \neq 0\}$. In particular, we are interested in partitions where $\text{supp } \kappa_i$ is a triangular block.

Analogous to non-fractional edge partitions, we can define contributions $e(\kappa_i) = \sum_{e \in E(G)} \kappa_i(e)$ and $\text{fc}(\kappa_i) = \sum_{f \in F(G)} \sum_{e \in f} \kappa_i(e) \frac{1}{|f|}$.

Lemma 4.1.

$$\begin{aligned} \sum_{i \in I} e(\kappa_i) &= |E(G)|. \\ \sum_{i \in I} \text{fc}(\kappa_i) &= |F(G)|. \end{aligned}$$

Define the quantity $q_i(\kappa_i) = \frac{e(\kappa_i)}{e(\kappa_i) - \text{fc}(\kappa_i)}$. Then, we have

Lemma 4.2. *Let $\kappa_i : E(G) \rightarrow [0, 1]$ be a fractional partition indexed over I . Let q Then,*

$$|E(G)| \leq \sup_{i \in I} q(\kappa_i)(|V(G)| - 2)$$

so it follows that $\exp(n, G) \leq q(\kappa_i)(n - 2)$.

Proof. Observe that

$$\begin{aligned} |E(G)| &= \sum_{i \in I} q(\kappa_i)(e(\kappa_i) - \text{fc}(\kappa_i)) \\ &\leq \sup_{i \in I} q(\kappa_i) \sum_{i \in I} (e(\kappa_i) - \text{fc}(\kappa_i)) \\ &= \sup_{i \in I} q(\kappa_i)(|E(G)| - |F(G)|) \\ &= \sup_{i \in I} q(\kappa_i)(|V(G)| - 2). \end{aligned}$$

□

To obtain bounds on the planar Turán numbers of I_m^ℓ , we use fractional partitions. Our proof for the bound on I_m^ℓ has the following outline.

1. Split $\mathcal{T}(G)$ into {blocks avoiding I_m } and {blocks containing I_m };
2. For any block avoiding I_m , use the bound given by 3.2;
3. For any block containing I_m , perform a fractional partition into copies of I_M with $M \geq m$.
*** make precise***
- 4.

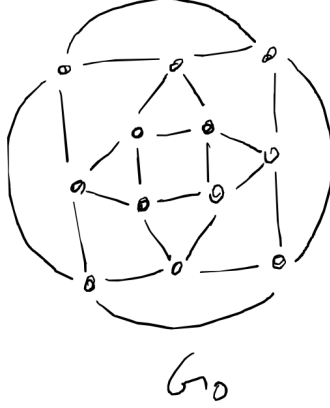


Figure 3: Building block G_0 for the extremal construction.

5 Extremal constructions

In the previous section, we bounded the planar Turán numbers of the triangular blocks I_4 , I_5 , I_6 , and I_7 . In this section, we will provide a construction which meets these bounds for infinitely many values of n .

Similar to the constructions in [1] and subsequent papers, our extremal construction takes the form of a graph G_0 on n vertices avoiding I_d which can be extended repeatedly to obtain larger graphs avoiding I_d . Specifically, let G_0 the graph in Figure 3. Then we define G_r inductively by extending G_{r-1} as shown in Figure 4. By inspection, the only triangular blocks present in G_r are copies of K_3 . Therefore, G_r avoids I_4 . By counting, we see that G_r has $12 + 20r$ vertices and $24 + 48r$ edges, so

$$\frac{\# \text{edges in } G_r}{(\# \text{vertices in } G_r) - 2} = \frac{24 + 48r}{10 + 20r} = \frac{12}{5}$$

Since G_r avoids I_4 , so this proves that the bound $\exp(n, I_4) \leq \frac{12}{5}(n - 2)$ given in Theorem 1.7 is tight whenever $n \equiv 12 \pmod{20}$.

To obtain extremal construction for I_5, I_6, I_7 as well as for I_5^1, I_5^2 , we replace the K_3 faces in G_r with plane triangulations in Figure ???. Let G_r^T be the graph obtained from G_r by replacing all K_3 faces with copies of T . Let t be the number of vertices in T , meaning $3t - 6$ is the number of edges in T . The number of T 's in G_r^T is $8 + 16r$ and this means G_r^T has

$$\begin{aligned} 12 + 20r + (t - 3)(8 + 16r) &= 2 + 16(t - \frac{7}{4})(r + \frac{1}{2}) \quad \text{vertices;} \\ (8 + 16r)(3t - 6) &= 48(t - 2)(r + \frac{1}{2}) \quad \text{edges.} \end{aligned}$$

Thus, we have the ratio

$$\frac{\# \text{edges in } G_r^T}{(\# \text{vertices in } G_r^T) - 2} = \frac{3(t - 2)}{t - \frac{7}{4}}.$$

In Figure ??, we gave a series of triangulations avoiding I_4, I_5, I_6, I_7 . Letting T be the triangulation from Figure ?? avoiding I_d , we find that G_r^T makes the inequality in Theorem 1.7 tight whenever

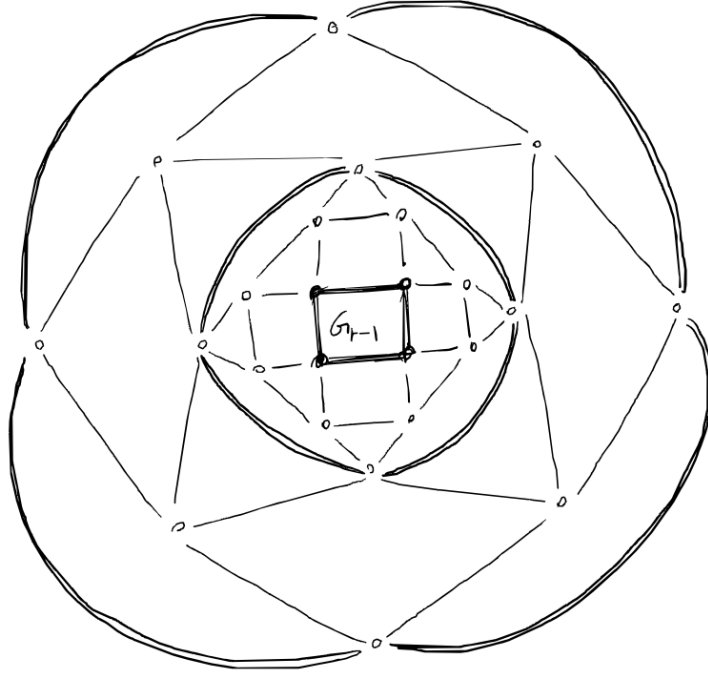


Figure 4: We extend G_{r-1} to G_r .

n is such that G_r^T can be constructed. The number of vertices in G_r^T is $(8t - 12) + (16t - 28)r$, so the bound is tight whenever $n \equiv 8t - 12 \pmod{16t - 28}$.

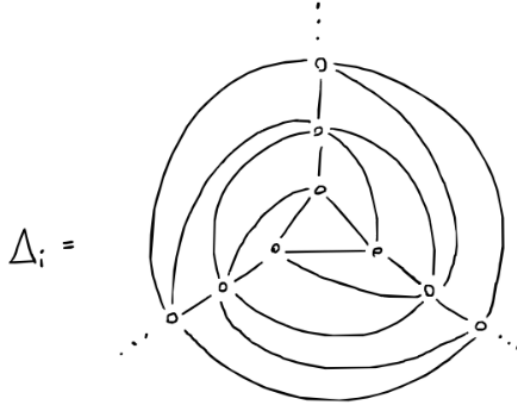


Figure 5: Triangulation Δ_n with all vertices having degree ≤ 6 .

6 $\exp(n, B) = 3n - 6$

The unique edge-maximal n vertex graphs are the complete graphs K_n with $\frac{n(n-1)}{2}$ vertices. On the other hand, the edge maximal graphs are plane triangulations with $3n - 6$ vertices and they are far from unique. This non-uniqueness means that there are graphs contained in some large plane triangulations but still avoided by infinitely many others. Graphs avoided by infinitely many plane triangulations necessarily have planar Turán number $3n - 6$ for infinitely many n .

In this section, we show that all but finitely many triangular blocks are avoided by some infinite family of plane triangulations. We begin by defining the two families of plane triangulations which achieve this.

Definition 6.1. For $n = 3m$, let Δ_n be the n vertex plane triangulation given in Figure 5. Let Δ_{3m+1} be $\Delta_{3m} + \{x\}$ where x is a new outer vertex connected to all three outer vertices of Δ_{3m} . Let Δ_{3m+2} be $\Delta_{3m+1} + \{y\}$ where y is a new outer vertex connected to all three outer vertices of Δ_{3m+1} .

The Δ_n graphs have two properties that are particularly advantageous: (1) all vertices in Δ_n have degree ≤ 6 and (2) any pair of vertices in Δ_n share at most 3 neighbors in common. This means the same properties must be true for any triangular block B with planar Turán number $\exp(n, B) < 3n - 6$.

Lemma 6.2. *Let B be a triangular block with planar Turán number $\exp(n, B) < 3n - 6$ for all n sufficiently large. Then,*

1. *B has maximal degree no greater than 6.*
2. *Any pair of vertices in B share at most 3 neighbors in common.*

Proof. Since Δ_n is a triangulation and $\exp(n, B) < 3n - 6$ for all n sufficiently large, so B must be a subgraph of Δ_n for all n sufficiently large. By inspection, every Δ_n satisfies (1) and (2) above.

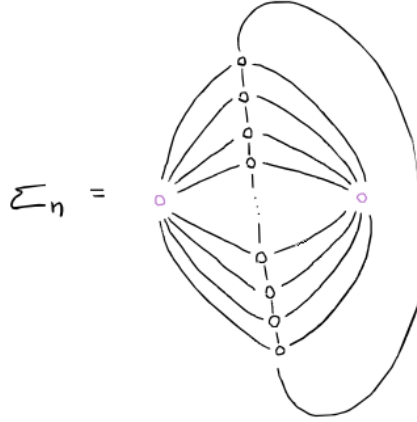


Figure 6: Triangulation Σ_n

Therefore, since B is a subgraph of some Δ_n , so B must satisfy (1) and (2) as well. $\square$

We now introduce another infinite family of plane triangulations with interesting properties.

Definition 6.3. For $n \geq 5$, let Σ_n be the n vertex plane triangulation given in Figure 6.

We think of Σ_n as having two *distinguished vertices* of degree $n - 2$ with all other vertices having degree 4. We now argue that Σ_{12} contains all triangular blocks with planar Turán number less than $3n - 6$.

Theorem 6.4. Let B be a triangular block with planar Turán Number $\text{ex}(n, B) < 3n - 6$ for all n sufficiently large. Then, B is a subgraph of Σ_{12} .

Proof. Suppose B is a triangular block with $\text{ex}(n, B) < 3n - 6$ which is not a subgraph of Σ_{12} . Since Σ_n is a triangulation, so B is contained in Σ_n for some n . Let n be the minimal value such that Σ_n contains B . Suppose for contradiction that $n \geq 13$.

Consider the $n - 2$ vertex cycle C consisting of all vertices except for the two distinguished vertices $\{v_1, v_2\}$. We first show that B contains every vertex of C . If there were a vertex $x \in C$ not contained in B . Then, choose some $y \in C$ with $y \sim x$ and contract the edge xy . This gives us the graph Σ_{n-1} which B must be a subgraph of; this contradicts our choice of n minimal. Therefore, $V(C) \subset V(B)$.

In Σ_n , every vertex in C is contained in four triangles containing either v_1 or v_2 ; consequently, the fact that every vertex in B is contained in a triangular face forces every vertex of C to be adjacent in B to at least one of v_1, v_2 .

This means B must contain at least $n - 2 \geq 11$ edges adjacent to one of the distinguished vertices. These edges can't all be adjacent to the same distinguished vertex since that would create a vertex

of degree > 6 contradicting Lemma 6.2 which says that B has maximum degree ≤ 6 . In order for B to contain vertices adjacent to both distinguished vertices, there must be some pair of vertices adjacent to both distinguished vertices. If this were not so, then B would not be a single triangular block.

Let a be the number of vertices of C adjacent only to v_1 , b the number adjacent only to v_2 , and c the number adjacent to both. We have $a + b + c = n - 2$ and the total number T of $\{v_1, v_2\}$ -to- C adjacencies (i.e., edges in $E(B)$ with one endpoint in $\{v_1, v_2\}$ and the other in C) equals

$$T = a + b + 2c = (n - 2) + c.$$

We have shown $c \geq 2$, hence $T \geq n$. Thus $E(B)$ contains at least n edges joining the distinguished vertices to vertices of C . By the pigeonhole principle, one of v_1, v_2 is incident in B to at least $\lceil \frac{T}{2} \rceil \geq \lceil \frac{n}{2} \rceil \geq 7$ such edges for $n \geq 13$. This contradicts Lemma 6.2, which bounds the maximum degree of B by 6.

Therefore, B must be a subgraph of Σ_{12} . □

Since Σ_{12} has only finitely many subgraphs, this theorem tells us that all but finitely many triangular blocks have planar Turán number equal to $3n - 6$. The rest of this section will be devoted to narrowing down this count to the 19 subgraphs in 1.8.

Remark 6.5 (3-connected graphs). A similar proof to Theorem 6.4 shows that, if G is a planar 3-connected graph with planar Turán number $\exp(G, n) < 3n - 6$, then G is a subgraph of Σ_{12} . Hence, all but finitely many 3-connected graphs have planar Turán number $3n - 6$.

Theorem 6.4 proves that only finitely many triangular blocks have planar Turán number less than $3n - 6$ for some n . However, there are a number of triangular blocks contained in Σ_{12} which are still avoided by Δ_n for all n . In particular, Lemma 6.2 (2) says that if B is contained in Δ_n , then any pair of vertices in B share at most 3 neighbors in common, so this is a property shared by any triangular block with planar Turán number $< 3n - 6$ for some n .

To obtain the list in Theorem 1.8, we consider other families of plane triangulations.

References

- [1] Chris Dowden. “Extremal C_4 -free/ C_5 -free planar graphs”. In: *J. Graph Theory* 83.3 (2016), pp. 213–230. ISSN: 0364-9024,1097-0118. DOI: 10.1002/jgt.21991. URL: <https://doi.org/10.1002/jgt.21991>.
- [2] Longfei Fang, Bing Wang, and Mingqing Zhai. “Planar Turán number of intersecting triangles”. In: *Discrete Math.* 345.5 (2022), Paper No. 112794, 10. ISSN: 0012-365X,1872-681X. DOI: 10.1016/j.disc.2021.112794. URL: <https://doi.org/10.1016/j.disc.2021.112794>.
- [3] Debarun Ghosh et al. “Planar Turán number of the 6-cycle”. In: *SIAM J. Discrete Math.* 36.3 (2022), pp. 2028–2050. ISSN: 0895-4801,1095-7146. DOI: 10.1137/21M140657X. URL: <https://doi.org/10.1137/21M140657X>.
- [4] Ervin Győri, Kitti Varga, and Xiutao Zhu. “A new construction for the planar Turán number of cycles”. In: *Graphs Combin.* 40.6 (2024), Paper No. 124, 5. ISSN: 0911-0119,1435-5914. DOI: 10.1007/s00373-024-02822-4. URL: <https://doi.org/10.1007/s00373-024-02822-4>.
- [5] Yongxin Lan, Yongtang Shi, and Zi-Xia Song. “Extremal H -free planar graphs”. In: *Electron. J. Combin.* 26.2 (2019), Paper No. 2.11, 17. ISSN: 1077-8926. DOI: 10.37236/8255. URL: <https://doi.org/10.37236/8255>.
- [6] Ruilin Shi, Zach Walsh, and Xingxing Yu. “Planar Turán number of the 7-cycle”. In: *European J. Combin.* 126 (2025), Paper No. 104134, 24. ISSN: 0195-6698,1095-9971. DOI: 10.1016/j.ejc.2025.104134. URL: <https://doi.org/10.1016/j.ejc.2025.104134>.
- [7] Chuanqi Xiao et al. “Planar Turán number of the Θ_6 ”. In: *Studia Sci. Math. Hungar.* 61.2 (2024), pp. 89–115. ISSN: 0081-6906,1588-2896.